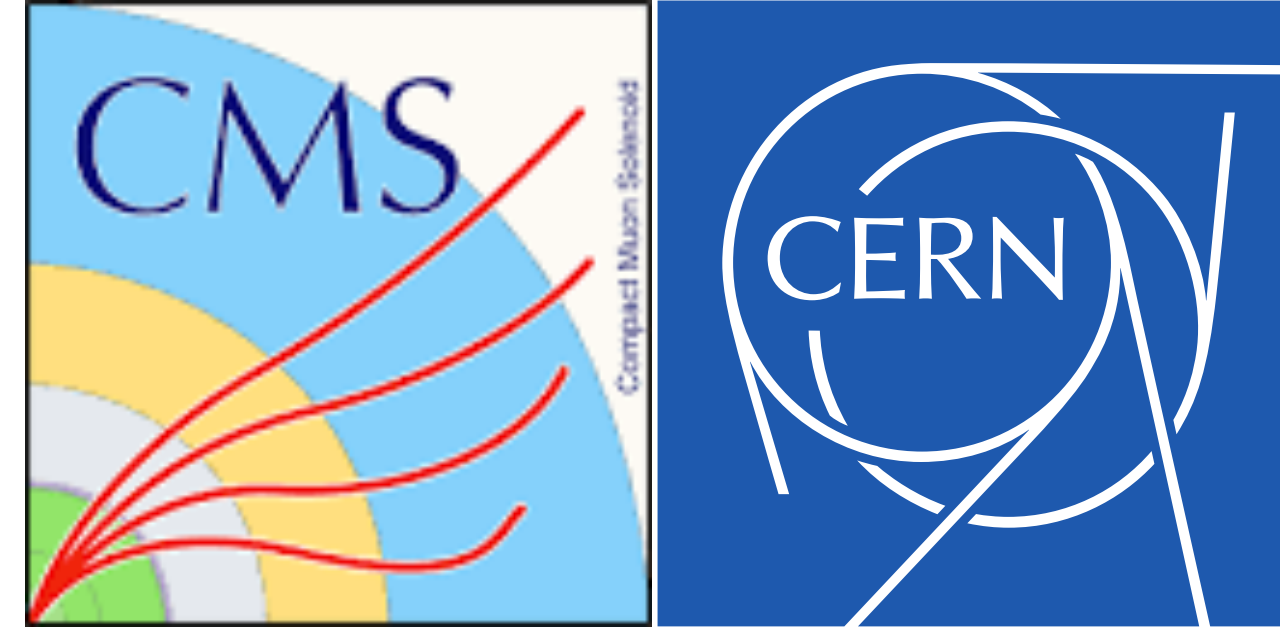




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DATA ANALYSIS

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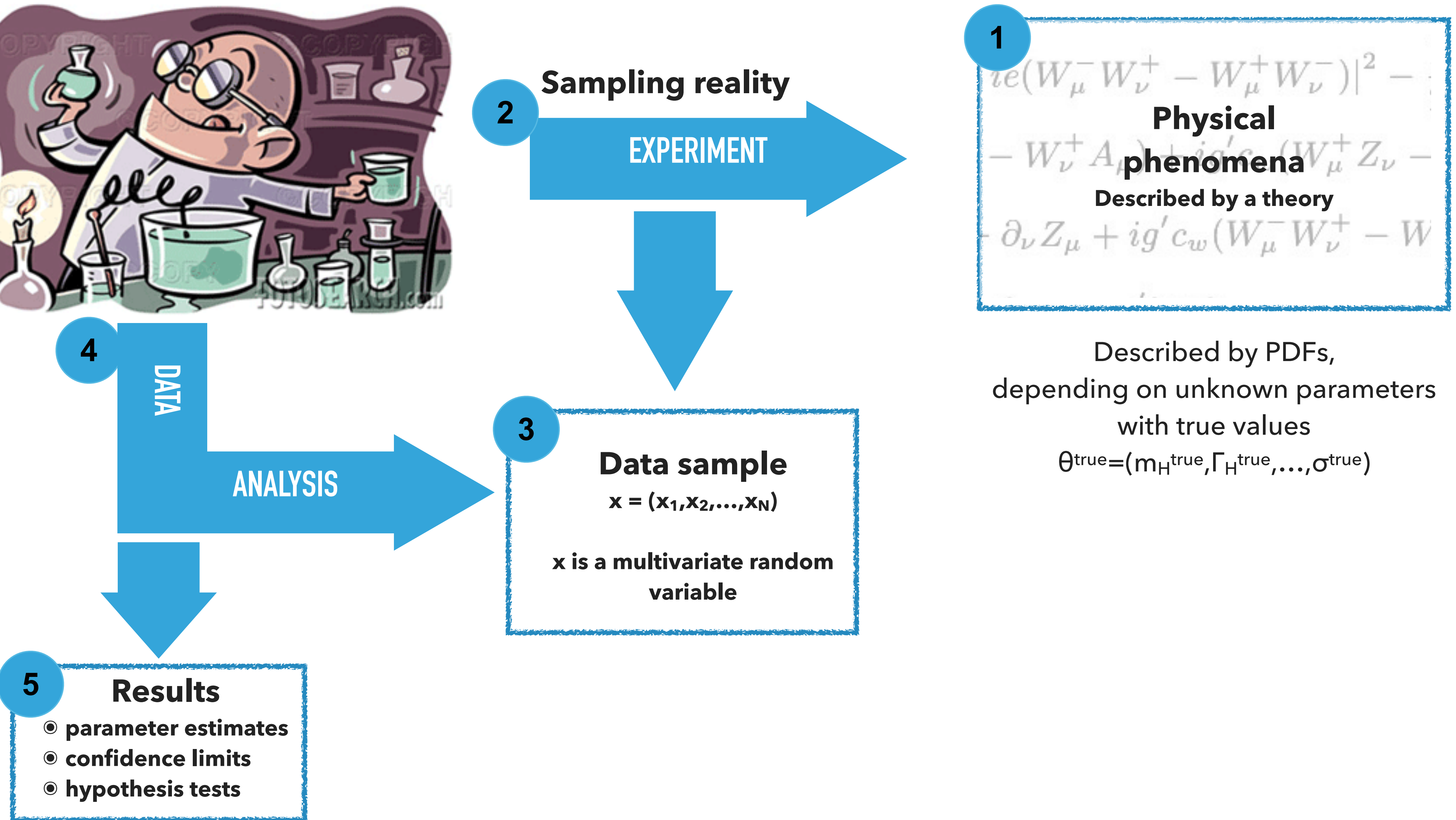
CERN School of Computing 2026, Santiago, Chile

- 1) Introduction to Data Analysis
- 2) Probability density functions and Monte Carlo methods
- 3) Parameter estimation
- 4) Confidence intervals
- 5) Hypothesis testing and p-value

PARAMETER ESTIMATION

GENERAL PICTURE REMINDER

4



- The parameters of a PDF are constants that characterise its shape:

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$$

- where x is measured data, and θ are parameters that we are trying to estimate (measure)
- Suppose we have a sample of observed values $\vec{x} = (x_1, x_2, \dots, x_n)$
- Our goal is to find some function of the data to estimate the parameter(s)
 - we write the **parameter estimator** with a hat $\hat{\theta}(\vec{x})$
 - we usually call the procedure of estimating parameter(s): **parameter fitting**

- Task: find the average height of all students in a university on the basis of an (honestly selected) sample of N students
- Some possible ways of getting the result:
 - 1) Add up all the heights and divide by N
 - 2) Add up the first 10 heights and divide by 10. Ignore the rest
 - 3) Add up all the heights and divide by $N-1$
 - 4) Throw away the data and give the answer as 1.8 m
 - 5) Multiply all the heights and take the N -th root
 - 6) Choose the most popular height (the mode)
 - 7) Add up the tallest and shortest height and divide by 2
 - 8) Add up the second, fourth, etc. and divide by $N/2$ for N even or by $(N-1)/2$ for N odd

● Consistent

- Estimate converges to the true value as amount of data increases

$$\hat{\theta} \xrightarrow{\text{more data}} \theta^{true}$$

● Unbiased

- Bias is the difference between expected value of the estimator and the true value of the parameter

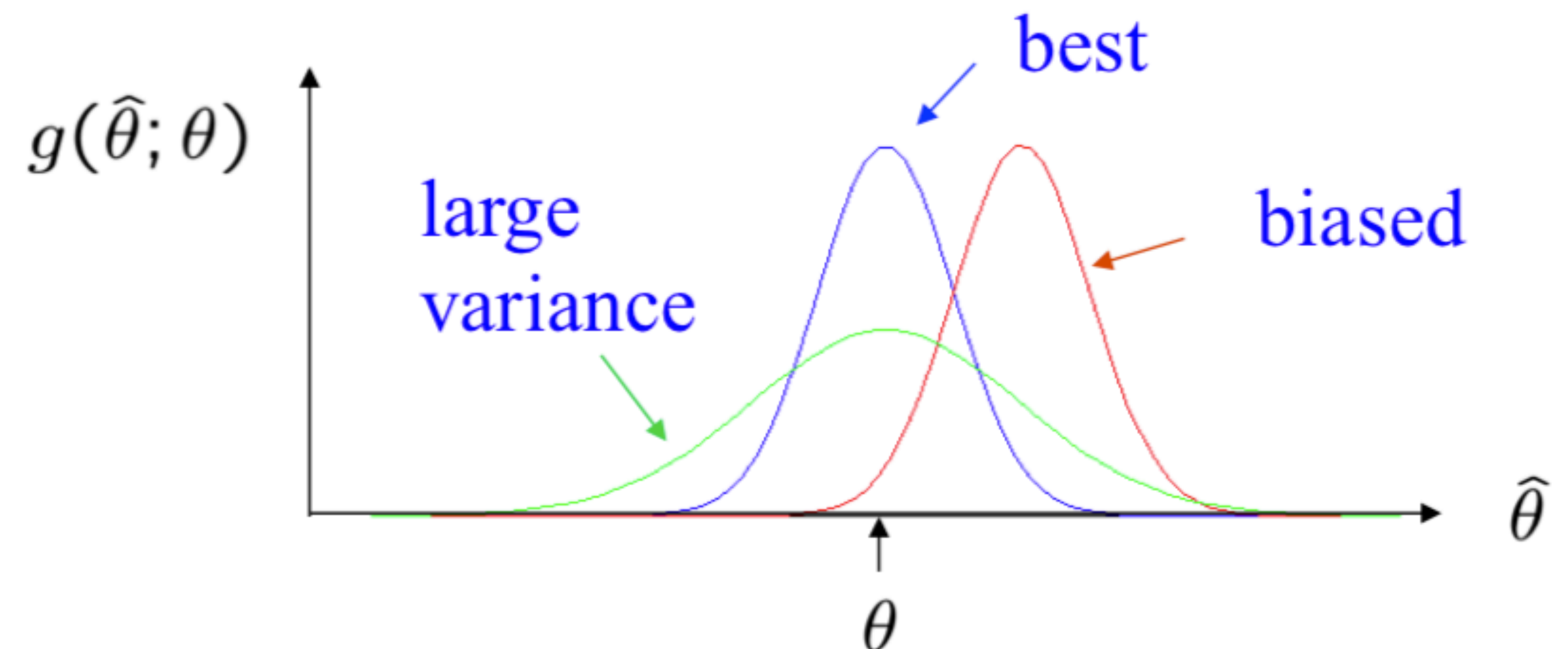
$$b = E(\hat{\theta}) - \theta^{true} = 0$$

● Efficient

- Its variance is small

● Robust

- Insensitive to departures from assumptions in the PDF



BONUS PROBLEM - 3

8

Quarks produced in high energy collisions will hadronize and form “jets” of particles. We call jets coming from the hadronization of b quarks “b-jets”. Algorithms to identify b-jets, referred to as b-tagging, will tag jets with a high probability to be b-jets. Their performance is characterized by two numbers:

1. The efficiency to tag real b-jets: $\varepsilon_b = P(\text{tag} \mid \text{b jet})$
2. The mistag rate to tag light flavour jets: $\varepsilon_{\text{mistag}} = P(\text{tag} \mid \text{light flavour jet})$

In an event with n_b true b-jets and n_{light} true light jets what is the probability to find n_{tag} tagged jets given ε_b and $\varepsilon_{\text{mistag}}$?

As example consider the process in which the Higgs boson is produced together with a top anti-top pairs, with the H decaying into a pair of b-jets, one of the top quarks decaying hadronically and the other semileptonically: $t\bar{t}H \rightarrow b\bar{b}l\bar{l} + bqq' + b\bar{b}$ (4b – jets + 2 light jets)

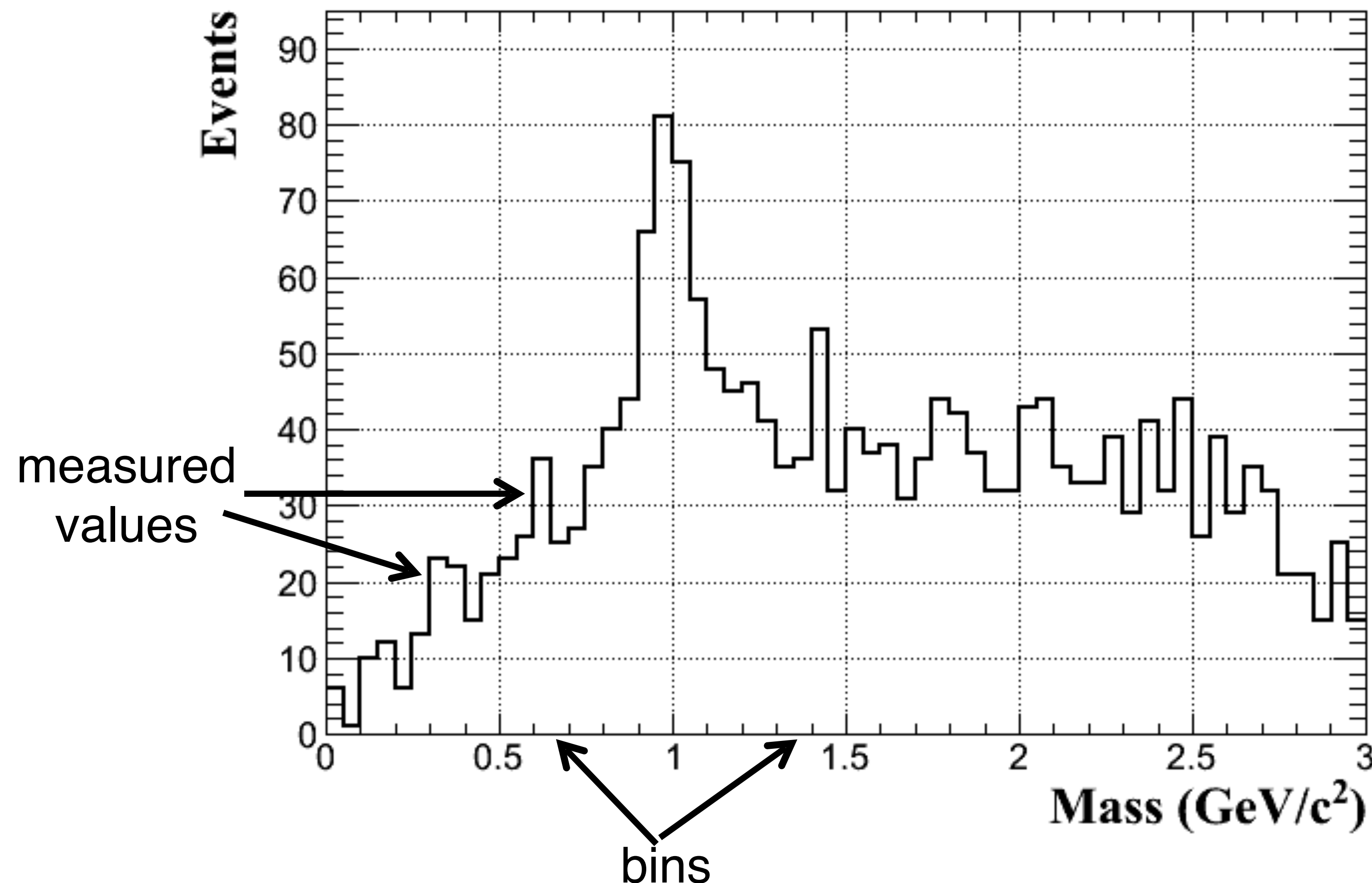
What is the probability to tag 2, 3, 4, 5 or 6 jets if $\varepsilon_b = 68\%$ and $\varepsilon_{\text{mistag}} = 1\%$

Hint! Let binomial distribution and python help you solve this one!

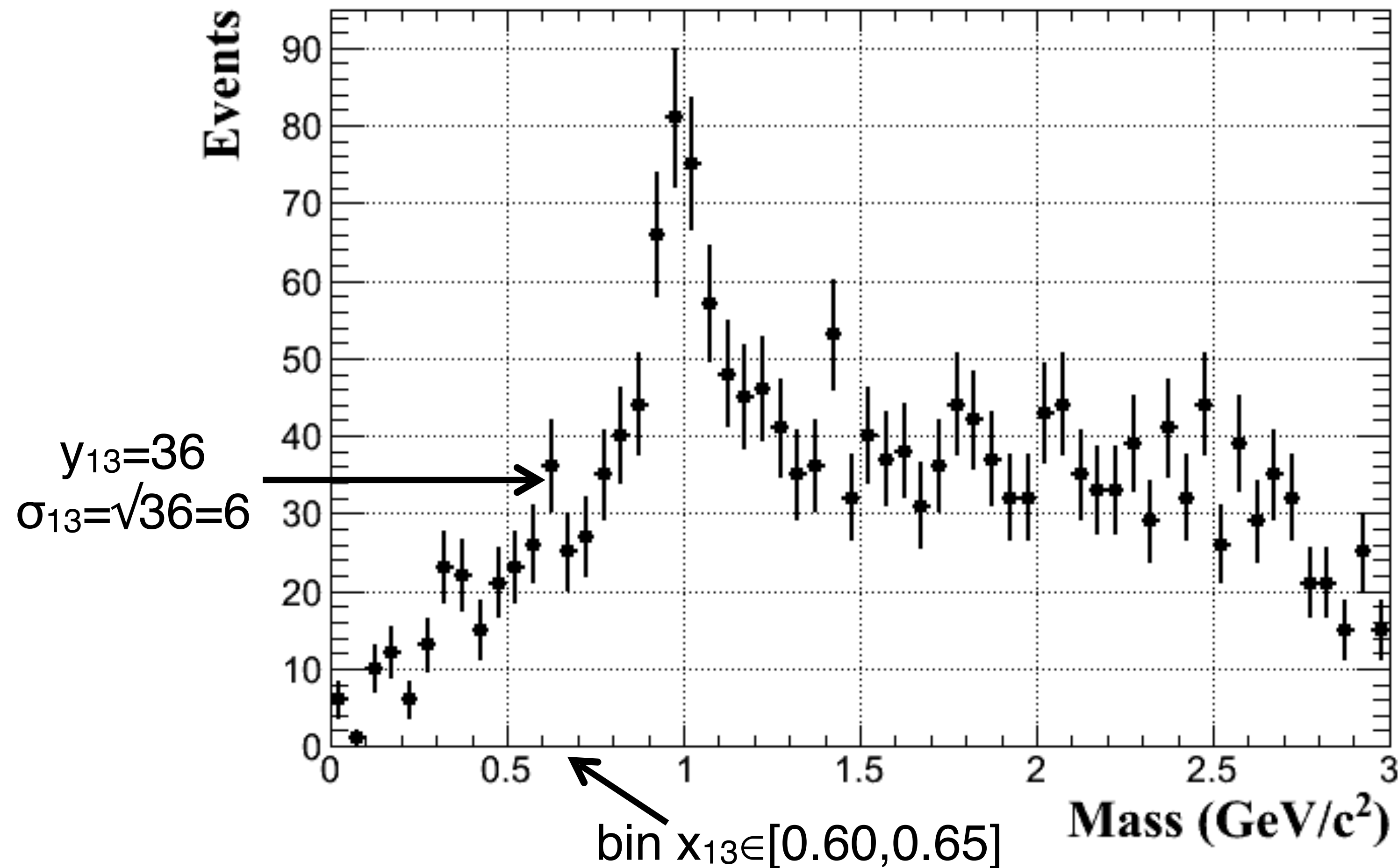
EXAMPLE IN HEP - HISTOGRAM FITTING

9

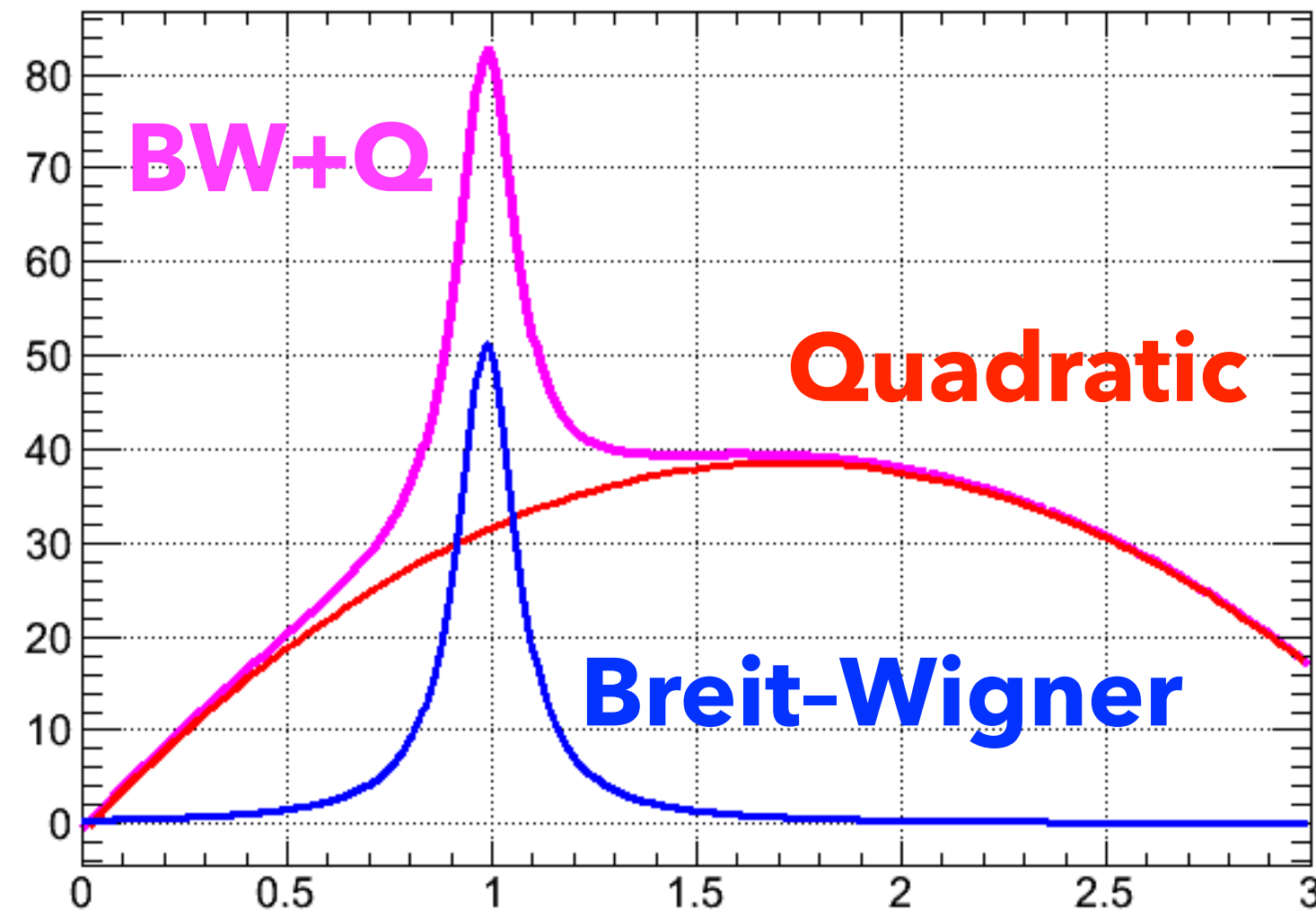
- In counting experiments we usually represent data in histograms
- In the following example we will study a particle mass histogram



- Measured values have statistical uncertainties so it is better to represent them with points and error bars
 - each bin has a Poisson uncertainty



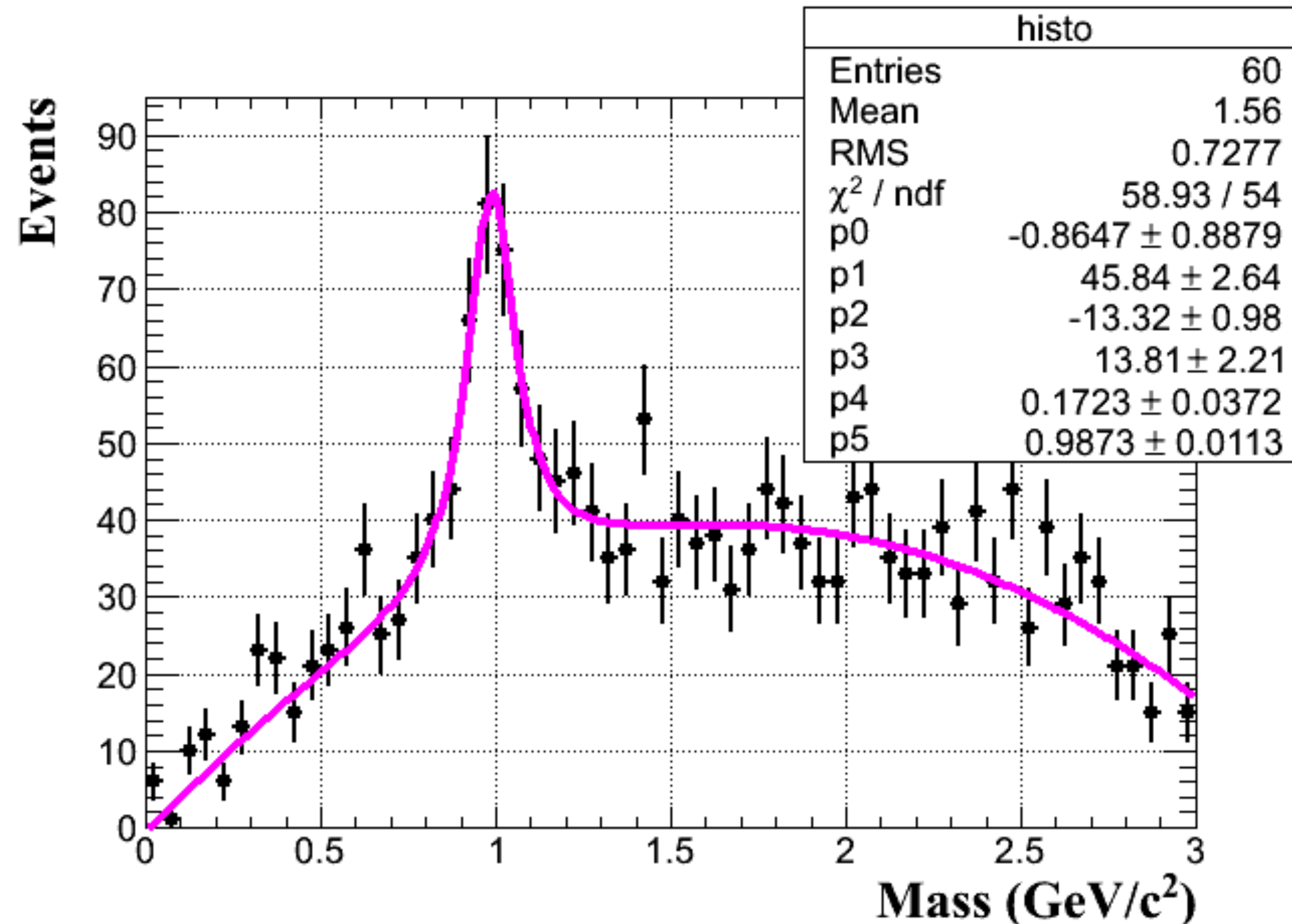
- Therefore we have
 - a set of precisely known values $\mathbf{x} = (x_1, \dots, x_N)$ - **histograms bins**
 - At each x_i
 - a measured value y_i - **number of events in a given bin**
 - a corresponding **error on measured value** σ_i
- We are missing a theoretical PDF $f(x_i; \theta^{true})$ with true parameters θ^{true} so we can calculate **parameter estimator** $\hat{\theta}$



$$BW(x; D, \Gamma, M) \approx \frac{D\Gamma}{(x^2 - M^2)^2 + 0.25\Gamma^2}$$

$$Q(x; A, B, C) = A + Bx + Cx^2$$

$$f(x_i, \theta^{true}) = f(x_i; D, \Gamma, M, A, B, C) = BW(x_i; D, \Gamma, M) + Q(x_i; A, B, C)$$



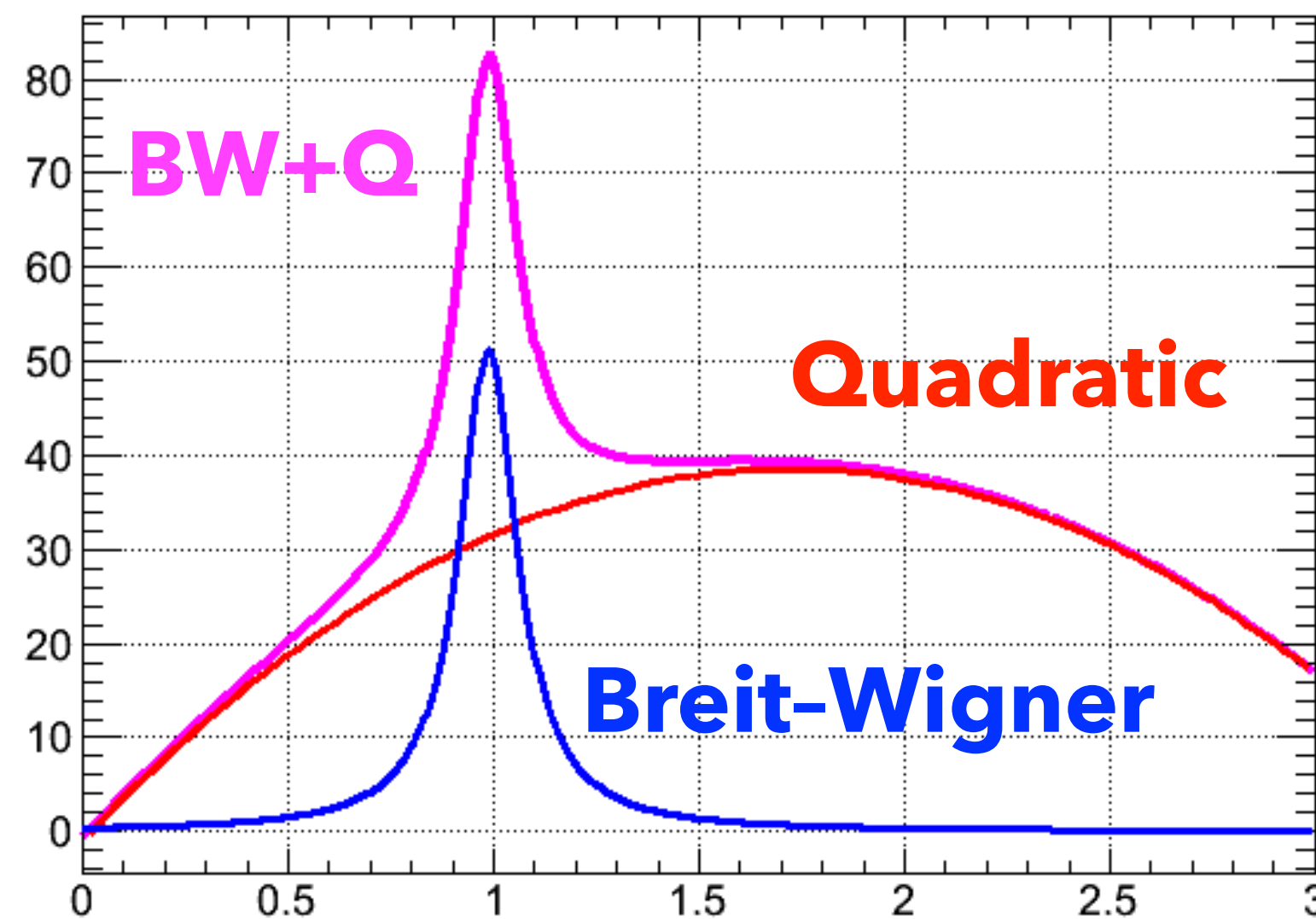
EXAMPLE IN HEP - HISTOGRAM FITTING

13

1

Physical phenomena

Described by a theory



2

Sampling reality

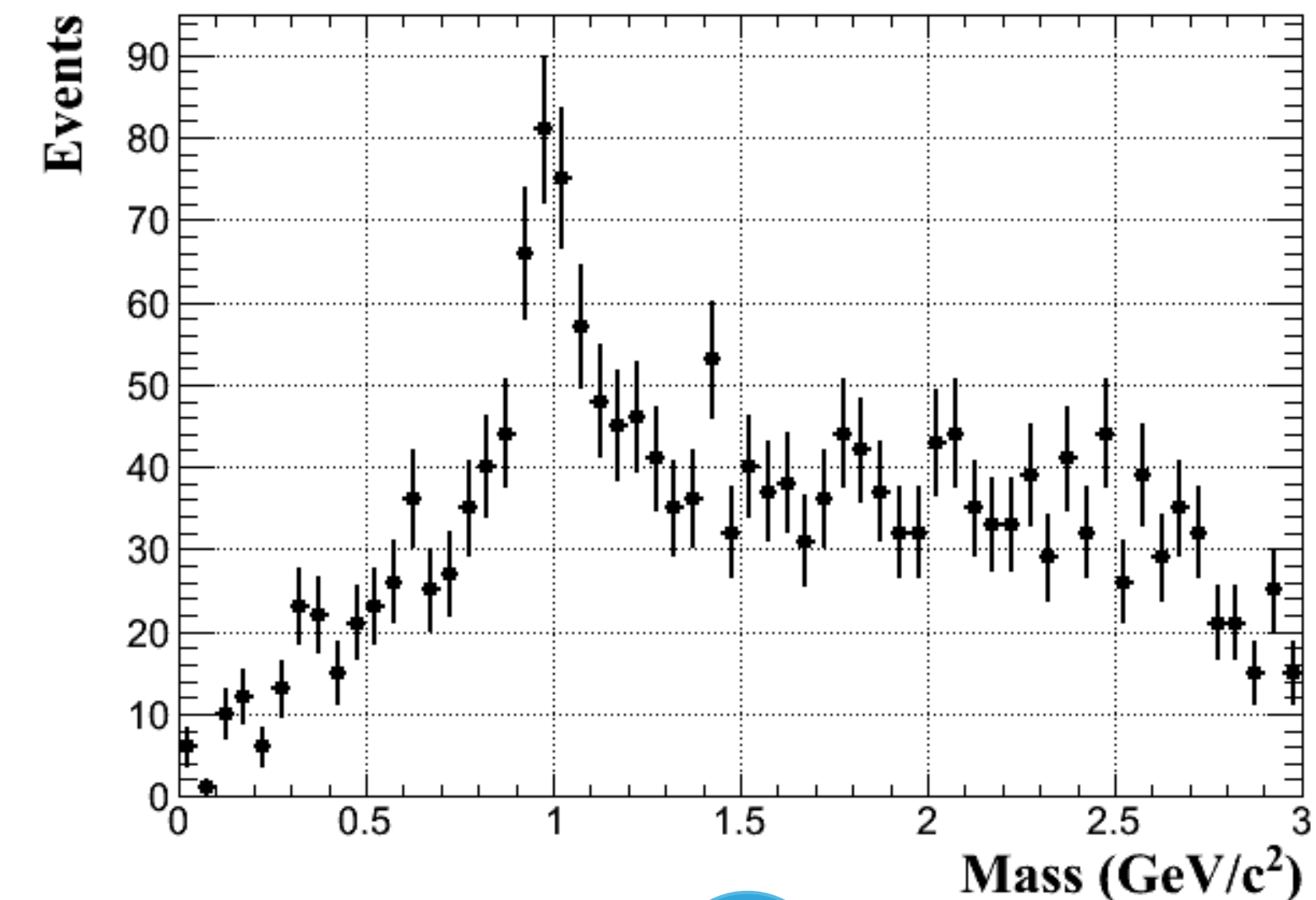
EXPERIMENT

3

Data sample

$\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$

$\mathbf{x}$ is a multivariate random variable



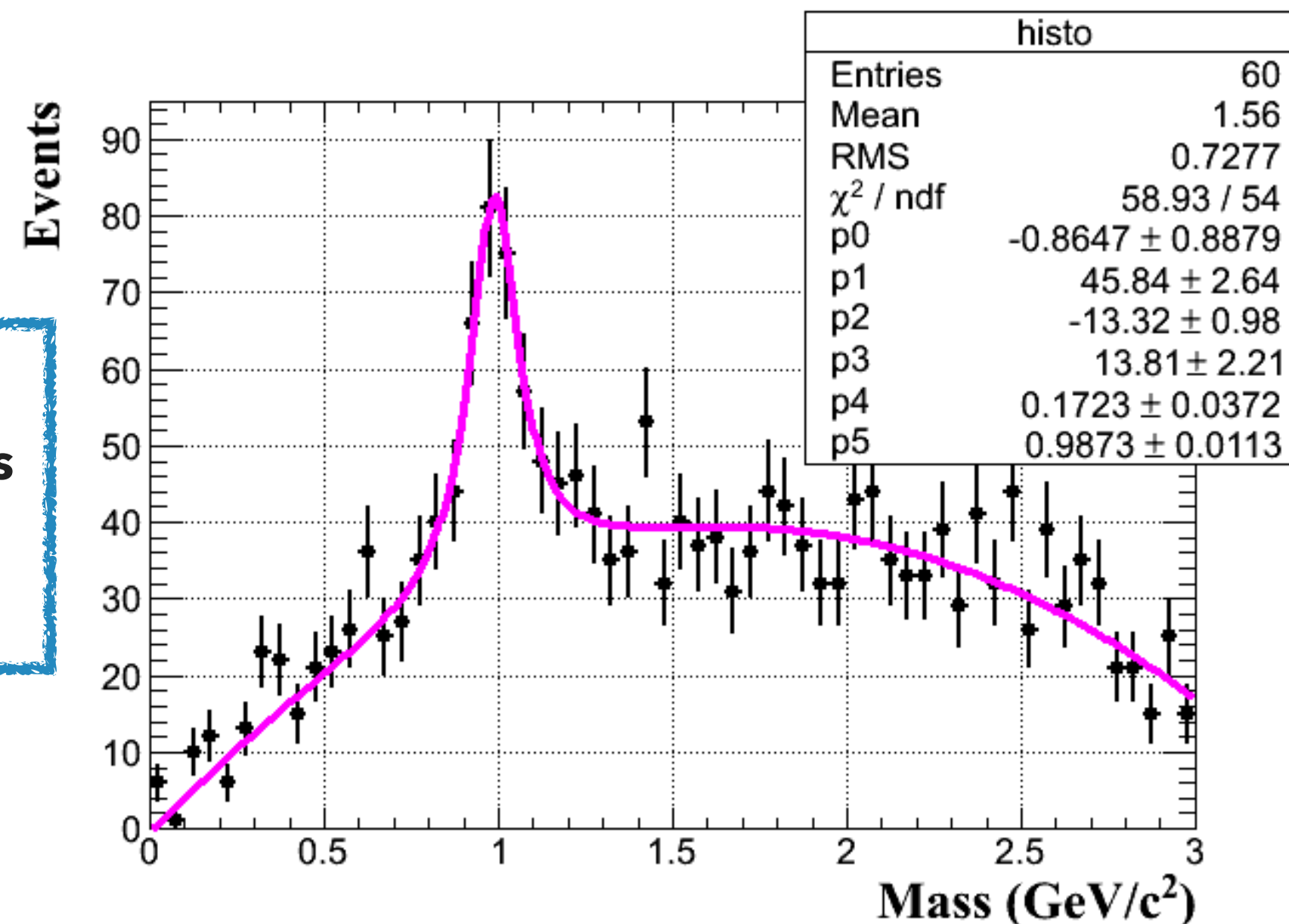
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DATA ANALYSIS

5

Results

- parameter estimates
- confidence limits
- hypothesis tests



- Be careful: **statistic** is not **statistics**!
- Any new random variable (f.g. T), defined as a function of a measured sample x is called a statistic $T = T(x_1, x_2, \dots, x_N)$
 - For example, the sample mean $\bar{x} = \frac{1}{N} \sum x_i$ is a statistic!
- A statistic used to estimate a parameter is called an **estimator**
 - For instance, the **sample mean** is a statistic and an estimator for the **population mean**, which is an unknown parameter
 - **Estimator** is a function of the data
 - **Estimate**, a value of estimator, is our “best” guess for the true value of parameter
- Some other example of statistics (plural of statistic!): sample median, variance, standard deviation, t-statistic, chi-square statistic, kurtosis, skewness, ...

THE MAXIMUM LIKELIHOOD METHOD

- Gives consistent and asymptotically unbiased estimators
- Widely used in practice

THE LEAST SQUARES (CHI-SQUARE) METHOD

- Gives consistent estimator
- Linear Chi-Square estimator is unbiased
- Frequently used in histogram fitting

- Assume that observations (events) are independent
 - With the PDF depending on parameters θ : $f(x_i; \theta)$
- The **probability that all N events will happen** is a product of all single events probabilities:
 - $P(x; \theta) = P(x_1; \theta)P(x_2; \theta) \cdots P(x_N; \theta) = \prod P(x_i; \theta)$
- When the variable **x is replaced by the observed** data x^{OBS} , then P is no longer a PDF
- It is usual to denote it by L and called $L(x^{\text{OBS}}; \theta)$ **the likelihood function**
 - Which is now a function of θ only $L(\theta) = P(x^{\text{OBS}}; \theta)$
- Often in the literature, it's convenient to keep X as a variable and continue to use notation $L(X; \theta)$

- The probability that all N independent events will happen is given by the likelihood function $L(x; \theta) = \prod f(x_i; \theta)$
- The principle of maximum likelihood (ML) says: **The maximum likelihood estimator $\hat{\theta}$ is the value of θ for which the likelihood is a maximum!**
- In words of R. J. Barlow: “You determine the value of θ that makes the probability of the actual results obtained, $\{x_1, \dots, x_N\}$, as large as it can possible be.”
- In practice it's easier to maximize the **log-likelihood function**
 $\ln L(x; \theta) = \sum \ln f(x_i; \theta)$
- For p parameters we get a set of p **likelihood equations**: $\frac{\partial \ln L(x; \theta)}{\partial \theta_j} = 0$
- It is often more convenient the **minimise $-\ln L$ or $-2\ln L$**

- Consider the lifetime pdf $f(t; \tau) = \frac{1}{\tau} e^{(-\frac{t}{\tau})}$
- Suppose we have **measured data** $t(t_1, \dots, t_N)$
- Our **likelihood function** is defined as $L(\tau) = \prod f(t_i; \tau)$
- The value of τ for which $L(\tau)$ is maximum also gives the maximum value of its **log-likelihood function** $\ln L(\tau) = \sum \ln f(t_i; \tau) = \sum (\ln \frac{1}{\tau} - \frac{t_i}{\tau})$
- Solving one likelihood equation $\frac{\partial \ln L(\tau)}{\partial \tau} = 0$ gives $\hat{\tau} = \frac{1}{N} \sum t_i$
- Try generating 100 Monte Carlo toys for $\tau = 1$ and estimating $\hat{\tau}$ using the ML method

- ML estimator is **consistent**
- ML estimate is approximately **unbiased** and **efficient** for large samples
 - Usually biased for small samples
- ML estimate is **invariant**
 - A transformation of parameter won't change the answer
 - Keep in mind that invariance comes at the cost of a bias!
- Extra care to be taken when the best value of parameters are near imposed limits
- **ML estimate is not the most likely value of parameter; it is the estimate that makes your data the most likely!**
- ML method can be used in the Bayesian approach where both θ and x are random variables
- We want to know the conditional PDF for θ given the data x :
$$p(\theta | x) = \frac{L(x | \theta)\pi(\theta)}{\int L(x | \theta')\pi(\theta')d\theta'}$$

- Suppose you have a set of precisely known (without error) values $x(x_1, \dots, x_N)$ with a corresponding set of measured values $y(y_1, \dots, y_N)$ with corresponding uncertainties $\sigma(\sigma_1, \dots, \sigma_N)$
 - For example x_i histogram mass bins with y_i events with Poissonian uncertainty σ_i
- Suppose you also know a function $f(x; \theta)$ which predicts the value of y_i for any x_i . It depends on an unknown parameter θ , which you are trying to determine.
 - In our example function $f(x; \theta)$ would be theoretical prediction for number of events at a given mass
- To find best estimate of θ we minimise the suitably weighted sum of squared differences between measured and predicted values, the so called “**least squares**” or “**chi-square**”:

$$\chi^2(\theta) = \sum_{i=1}^N \frac{(y_i - f(x_i; \theta))^2}{\sigma_i^2}$$

● Estimator is found by finding the value which minimises $\chi^2 : \frac{\partial \chi^2}{\partial \theta} = 0$

● The quantity $\chi^2 = \sum_{i=1}^N \frac{(y_i^{data} - y_i^{ideal})^2}{(expected\ error)^2}$ gives information about the fit quality

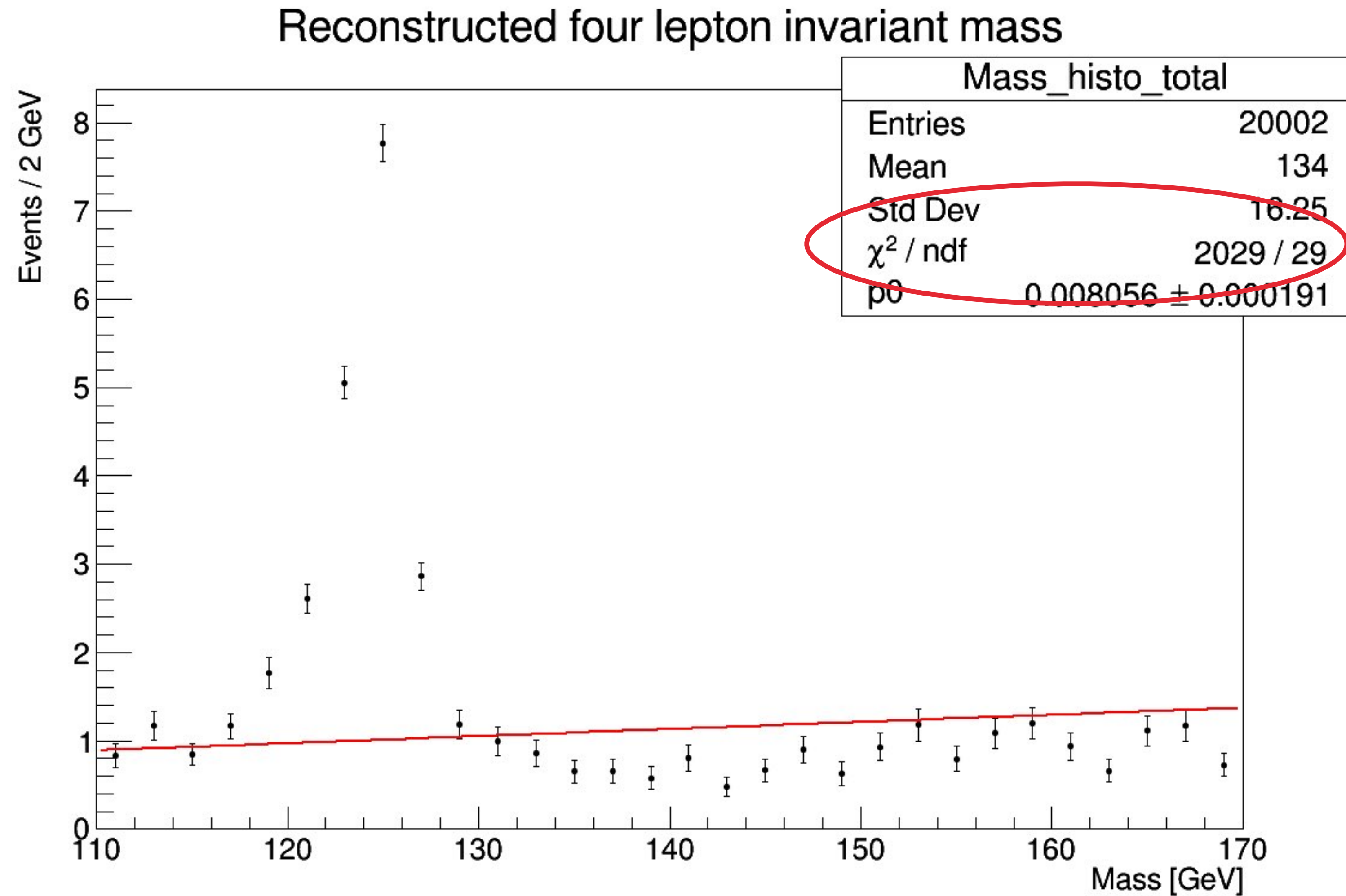
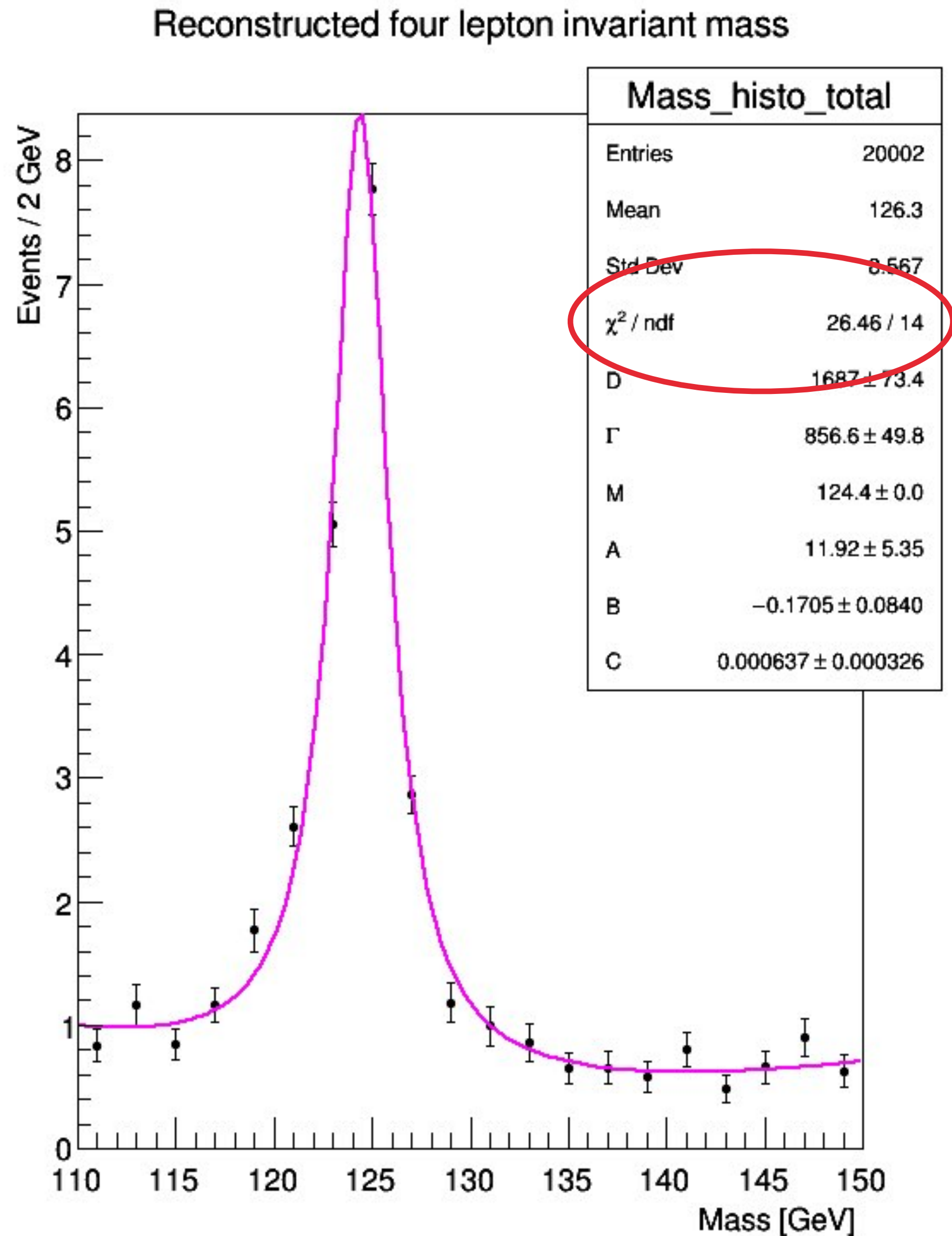
small χ^2	large χ^2
good fit	bad fit (bad model)
overestimated errors	underestimated errors

● Since $\langle \chi^2 \rangle = N$, easy way to estimate the fit quality is to check if $\frac{\chi^2}{N.D.O.F} \approx 1$, N.D.O.F is calculated as (N - free parameters)



CHI-SQUARE FIT TEST - EXAMPLE

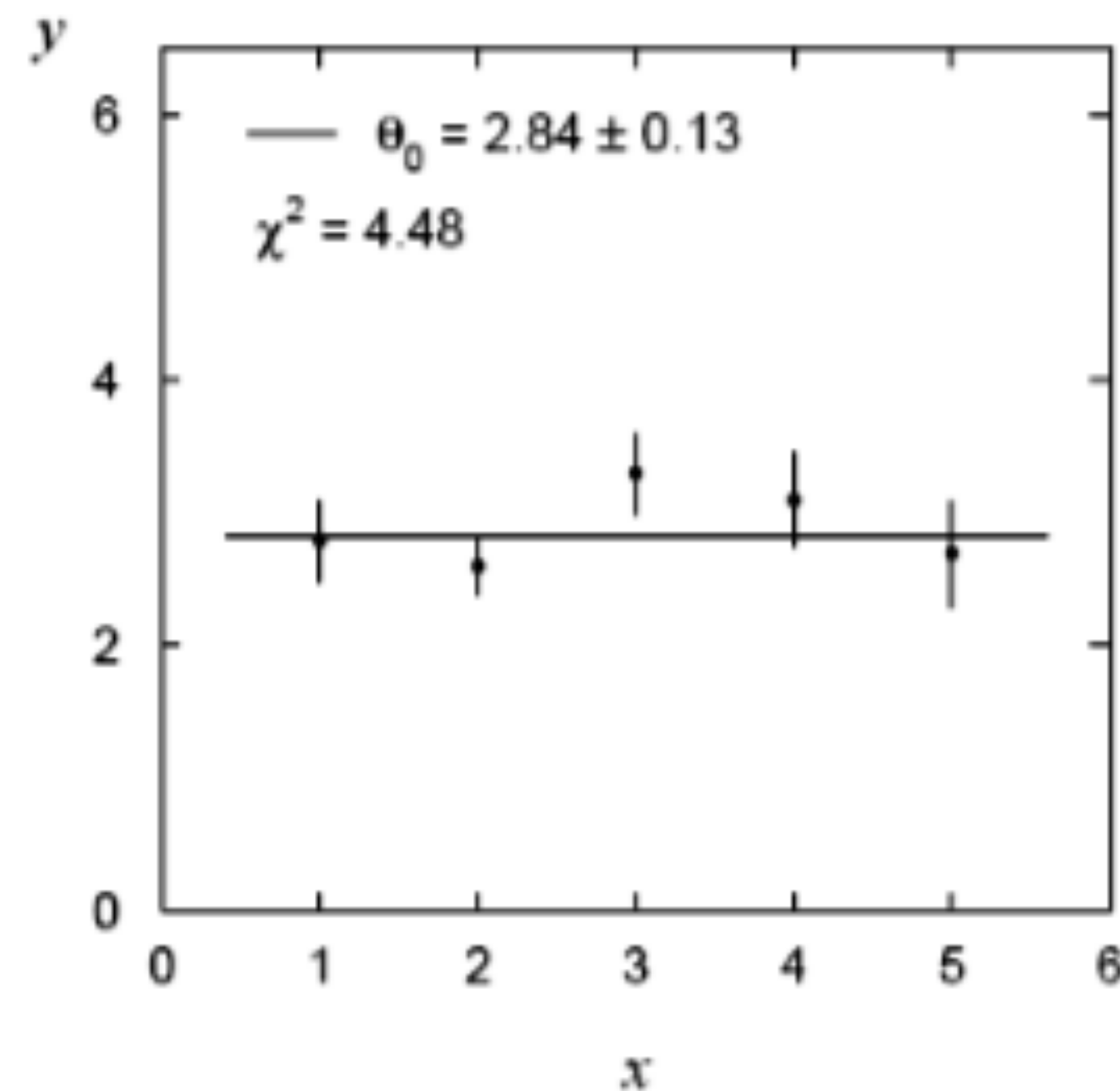
23



- LS has particularly desirable properties if $f(x; \theta)$ is a linear function of θ :

$$f(x; \theta) = \sum_{j=1}^m a_j(x) \theta_j, \text{ where } a_j(x) \text{ are linearly independent functions of } x$$

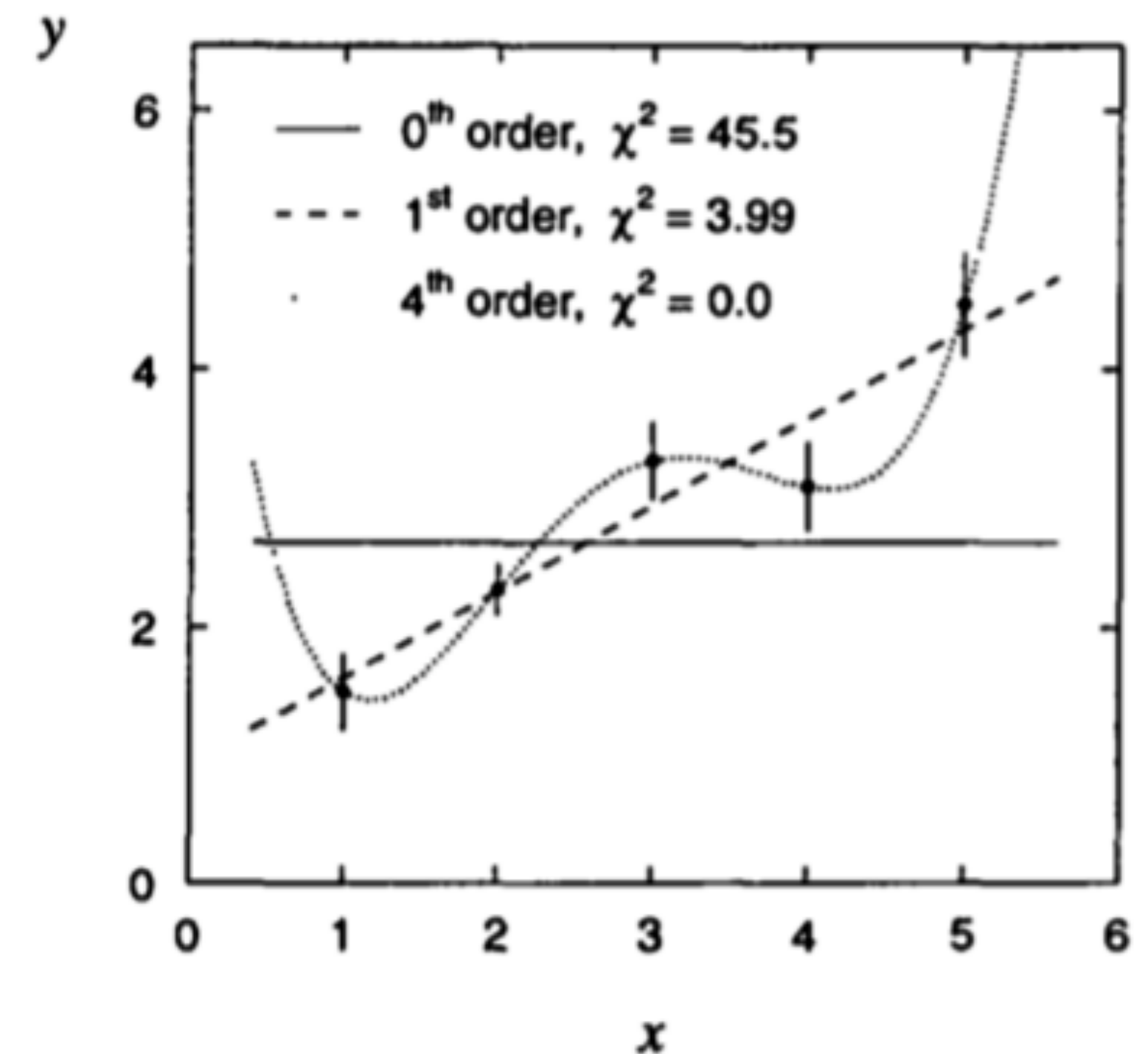
- estimators and their variances can be found analytically
- the estimators have zero bias and minimum variance



- Assume we measure 5 values of a quantity y , measured with errors σ_y at different values of x

- For the fit function we try polynomial of order m :
$$f(x; \theta) = \sum_{j=0}^m x^j \theta_j$$

- 0-th order: the weighted average
- 1-st order: a very good description
- 4-th order: equal number of parameters as points
- For Gaussian distributed y LS = ML!





- In the usual maximum likelihood method
 - Parameter relevant to the **shapes** of distributions are determined
 - Absolute **normalization** is **equal** to the **observed** number of events
- Sometimes in the experiment number of measurements N is not fixed but a random variable distributed according to the Poissonian with a mean ν
 - $x = (x_1, \dots, x_N)$
- The **extended likelihood function** is $L(\nu; \theta) = \frac{\nu^N e^{-\nu}}{N!} \prod f(x_i; \theta)$
- If theory gives $\nu = \nu(\theta)$ then the extended log-likelihood function is defined as
- $\ln L(\theta) = -\nu(\theta) + \sum_{i=1}^N \ln(\nu(\theta) f(x_i; \theta)) + C$

- From the vector of measurements $x = (x_1, \dots, x_N)$ we want to estimate number of signal events (s), number of background events (b) and a vector of parameters $\theta = (\theta_1, \dots, \theta_p)$

- Extended likelihood function is

$$L(x; s, b, \theta) = \frac{(s + b)^N e^{-(s+b)}}{N!} \prod_{i=1}^N \left(\frac{s}{s + b} f_s(x_i; \theta) + \frac{b}{s + b} f_b(x_i; \theta) \right)$$



- To obtain parameter estimates for s , b , and $\theta = (\theta_1, \dots, \theta_p)$ we maximise

$$\ln L(x; s, b, \theta) = -s - b + \sum_{i=1}^N \ln \left(\frac{s}{s + b} f_s(x_i; \theta) + \frac{b}{s + b} f_b(x_i; \theta) \right) - \ln(N!)$$

- Likelihood function (L) is constructed by replacing the variable x by the observed data in a product of single events probabilities
- Maximising (minimising) the $\ln L$ ($-2 \ln L$) function gives the parameter estimate $\hat{\theta}_{ML}$
- $\hat{\theta}_{ML}$ does not mean that the estimate is the “most likely” value of θ , it is the value that makes your data most likely
- ML estimate is unbiased and efficient for large samples, be careful if you want to use it for small samples
- ML can be used to fit binned data
- ML can be extended to deal with the case where the number of expected events is not a fixed number but a random number